

AI for IoT: Practical Works 1 & 2

Fire Alarm Detection

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1 TP1: Fire Alarm Detection

Objective: Build a machine-learning system to detect fire alarms from IoT sensor data.

Dataset and EDA: The dataset contains 62,630 samples, 13 input features, and the target `Fire_Alarm`. No missing values were found. EDA included dataset inspection and correlation analysis.

Preprocessing: Unnecessary columns were removed. The data was split into 80% training and 20% testing. `StandardScaler` was fitted on the training data to prevent data leakage.

Results:

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	98.93%	99.27%	99.23%	99.25%
XGBoost	100%	100%	100%	100%

Recall is important because false negatives represent fire events that the model fails to detect.

2 TP2: Optimization and Deployment

Pipelines combined preprocessing and classification. Model size, training time, and inference time were measured in Google Colab.

Model	Size (KB)	Training (s)	Inference (s)
LR-Pipeline	1.69	0.728	0.0159
XGB-Pipeline	116.77	3.272	0.0479

Average single-sample inference time was 0.001269 ms for Logistic Regression and 0.003820 ms for XGBoost in Colab.

3 Conclusion

Logistic Regression is smaller and faster, making it a promising choice for ESP32 deployment. XGBoost achieved higher test accuracy but requires more resources. Both models require validation on the actual hardware before deployment.